

Object-Oriented Programming, Spring 2026
Assignment-1

Course Code: **CSC104**
Course Instructor: **Engr. Majid Kaleem**

Submission Deadline: **25/03/26**
Max. Marks: **04**

Question 1:

- a) Define the four pillars of OOP (Abstraction, Encapsulation, Inheritance, Polymorphism) in your own words. Provide one real-world analogy for each.
- b) Explain the role of access modifiers (public, private, protected, and default) in Java. How do they support encapsulation?
- c) Compare Encapsulation vs. Abstraction with examples.

Question 2:

You are designing a simple University Management System. The system should manage Students and Courses.

- a) Create a base class **Person** with attributes **name** and **age**.
- b) Use appropriate access modifiers for attributes.
- c) Provide **getter** and **setter** methods.
- d) Create a derived class **Student** that inherits from **Person**.
- e) Add attributes: **studentId** and **department**.
- f) Create a class **Course** with attributes **courseCode**, **courseName**, and **creditHours**.
- g) Apply encapsulation using **private** attributes and **public** methods.
- h) Demonstrate abstraction by creating an abstract class **Evaluation** with an abstract method **calculateGrade()**.
- i) Implement this in a subclass **ExamEvaluation**.

NOTE: Declare the above using Java syntax; an actual program in Java is not required.

G😊d Luck!